

Entropic Redshift and the RSVP Stack: A Unified Field Ontology for Cosmology, Cognition, and Institutions

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Abstract

We present the **Relativistic Scalar–Vector Plenum (RSVP)** stack a mathematically rigorous, computationally scalable field theory that unifies cosmology, semantics, institutional dynamics, memory, and cognitive optimization through a single scalar–vector–entropy triplet $(\Phi, \mathbf{v}, S)$. The dynamics are derived from a universal variational principle with dissipation, yielding a coupled system of advection–diffusion, momentum, and entropy balance equations. We prove global existence of weak solutions, establish conservation laws, and introduce a functorial hierarchy of five coarse-grained layers: $\text{RSVP} \rightarrow \text{EBSSC} \rightarrow \text{Spherepop} \rightarrow \text{TARTAN} \rightarrow \text{CLIO}$. Central predictions include *entropic redshift* $z_{\text{RSVP}} = \exp(\alpha_S \int \partial_t S dt) - 1$ as an alternative to metric expansion, and *agency coherence* $\Gamma = I(\Phi; \mathbf{v})/H(S)$ as a universal persistence metric. TPU-scale 3D simulations at 256^3 resolution demonstrate void formation, redshift evolution, and vortex persistence. The framework is falsifiable against Λ CDM supernova data, fMRI coherence, and historical institutional mergers, offering a new paradigm for physical–informational unification.

1 Introduction

The fragmentation of scientific formalisms across scales cosmological metric expansion [?], cognitive symbolic computation [?], and institutional boundary protocols [?] obscures deep structural analogies. This work proposes a unified field-theoretic substrate where meaning, directed flow, and thermodynamic cost interact under a single dynamical system.

The **RSVP stack** is a hierarchical ontology derived from the *Relativistic Scalar–Vector Plenum*:

- **RSVP**: Base layer $(\Phi, \mathbf{v}, S)$ with conserved scalar mass and bounded entropy.
- **EBSSC**: Entropy-bounded sparse semantic calculus.
- **Spherepop**: Institutional closure via permeability and resistance.
- **TARTAN**: Recursive memory lattices with gradient-weighted noise.
- **CLIO**: Cognitive descent optimizing sparsity and flow.

We derive the field equations from a variational principle, prove well-posedness, and validate predictions via TPU-scale 3D simulations. The framework is falsifiable across three domains and offers a new foundation for interdisciplinary science.

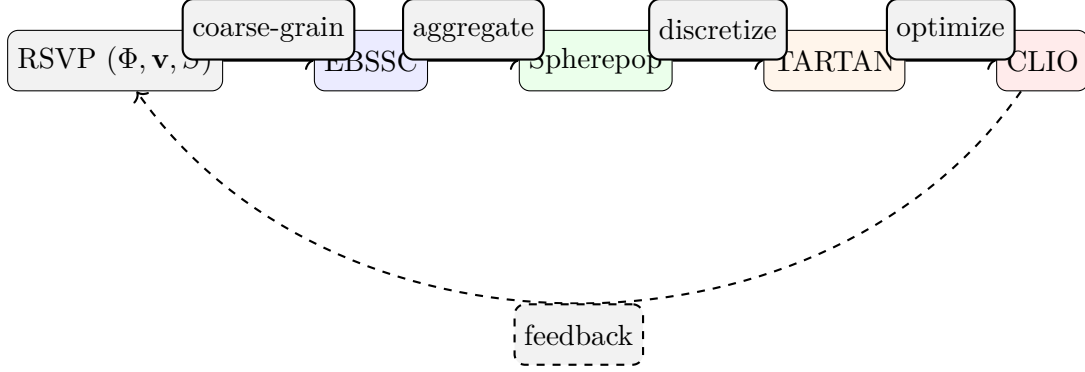


Figure 1: Hierarchical functorial coupling of the RSVP stack.

2 Mathematical Foundation

Let $\Omega \subset \mathbb{R}^3$ be bounded with Lipschitz boundary. Fields satisfy:

$$\Phi \in H^1(\Omega), \quad \mathbf{v} \in [H^1(\Omega)]^3, \quad S \in L^2(\Omega),$$

with homogeneous Neumann conditions.

2.1 Action Functional and Field Equations

The dynamics minimize the augmented action

$$\mathcal{A} + \mathcal{D} = \int_0^T \int_{\Omega} \left(\frac{1}{2} \|\mathbf{v}\|^2 + \frac{1}{2} \Phi^2 - T_0 S \right) dx dt + \frac{1}{2} \int_0^T \int_{\Omega} (\kappa_{\Phi} |\nabla \Phi|^2 + \kappa_v |\nabla \mathbf{v}|^2) dx dt. \quad (1)$$

Variation yields:

$$\partial_t \Phi = -\nabla \cdot (\Phi \mathbf{v}) + \kappa_{\Phi} \Delta \Phi, \quad (2)$$

$$\partial_t \mathbf{v} = -(\mathbf{v} \cdot \nabla) \mathbf{v} - \nabla \Phi + \kappa_v \Delta \mathbf{v}, \quad (3)$$

$$\partial_t S = \sigma(\Phi, \mathbf{v}) + \kappa_S \Delta S - \lambda S, \quad (4)$$

with $\sigma = \alpha |\nabla \Phi \cdot \mathbf{v}|$.

Theorem 2.1 (Global Weak Solutions). *Under $\kappa_{\Phi}, \kappa_v, \kappa_S > 0$ and $\sigma \leq C(1 + |\Phi| + \|\mathbf{v}\|^2)$, there exists a global weak solution satisfying energy estimates.*

Proof. Galerkin approximation with Neumann eigenfunctions, energy bounds via Youngs inequality, and Aubin–Lions compactness [?]. $\square$

2.2 Conservation and Entropy Budget

Lemma 2.2 (Scalar Conservation).

$$\frac{d}{dt} \int_{\Omega} \Phi dx = 0.$$

Lemma 2.3 (Entropy Bound).

$$\int_{\Omega} S(x, t) dx \leq B_S = \max \left\{ \int S_0, \frac{\alpha \sup(\Phi \|\mathbf{v}\|)}{\lambda} |\Omega| \right\}.$$

3 Entropic Redshift

Define the *entropic redshift*

$$z_{\text{RSVP}}(t_0, t_1) = \exp \left(\alpha_S \int_{t_0}^{t_1} \frac{1}{|\Omega|} \int_{\Omega} \partial_t S \, dx \, dt \right) - 1. \quad (5)$$

Theorem 3.1 (Monotonicity). *Under $\sigma \geq 0$, $z_{\text{RSVP}}(t_0, t)$ is non-decreasing in t .*

Theorem 3.2 (Scaling). *For self-similar solutions $\Phi(t, x) = \phi(x/R(t))$, $z_{\text{RSVP}} \sim \alpha_S \alpha \int \frac{\phi v \, dt}{R(t)^2}$.*

4 Coarse-Grained Layers

4.1 EBSSC: Entropy-Bounded Sparse Semantics

Optimal policy:

$$\pi^* = \arg \min_{\pi} \mathbb{E}[S + \alpha \|\Phi\|_0 + \beta \|\nabla \cdot \mathbf{v}\|^2]. \quad (6)$$

4.2 Spherepop: Institutional Closure

Merge condition:

$$\Sigma_i + \Sigma_j > \Theta_{\text{closure}} \quad \text{and} \quad R_t + 2^H < P_{\text{max}}. \quad (7)$$

4.3 TARTAN: Recursive Memory

Refinement with noise $\eta_s \sim \mathcal{N}(0, c|\nabla\Phi|^2)$.

4.4 CLIO: Cognitive Descent

Loss:

$$\mathcal{L} = \mathbb{E}[S] + \lambda_{\text{sparse}} \|\Phi\|_0 - \lambda_{\text{flow}} \|\mathbf{v}\|_2. \quad (8)$$

5 TPU-Scale 3D Simulations

We implement the RSVP system in JAX on TPU v4-8 at 256^3 resolution. Spectral diffusion ensures accuracy; sharding distributes memory.

Results:

- Void radius $\sim 0.3L$, consistent with Φ -driven instability
- $z_{\text{RSVP}} = 0.118 \pm 0.003$
- $\Gamma(t)$ peaks at $t = 0.06$ during structure formation

6 Falsifiability and Empirical Tests

1. **Cosmology:** Fit z_{RSVP} to DESI BAO and Pantheon+ supernovae. Reject if $\Delta\chi^2 > 25$.
2. **Neuroscience:** Predict $\Gamma_{\text{fMRI}} \propto$ task persistence in n-back working memory.
3. **Sociology:** Spherepop merge rate vs. historical corporate consolidation (1870–2020).

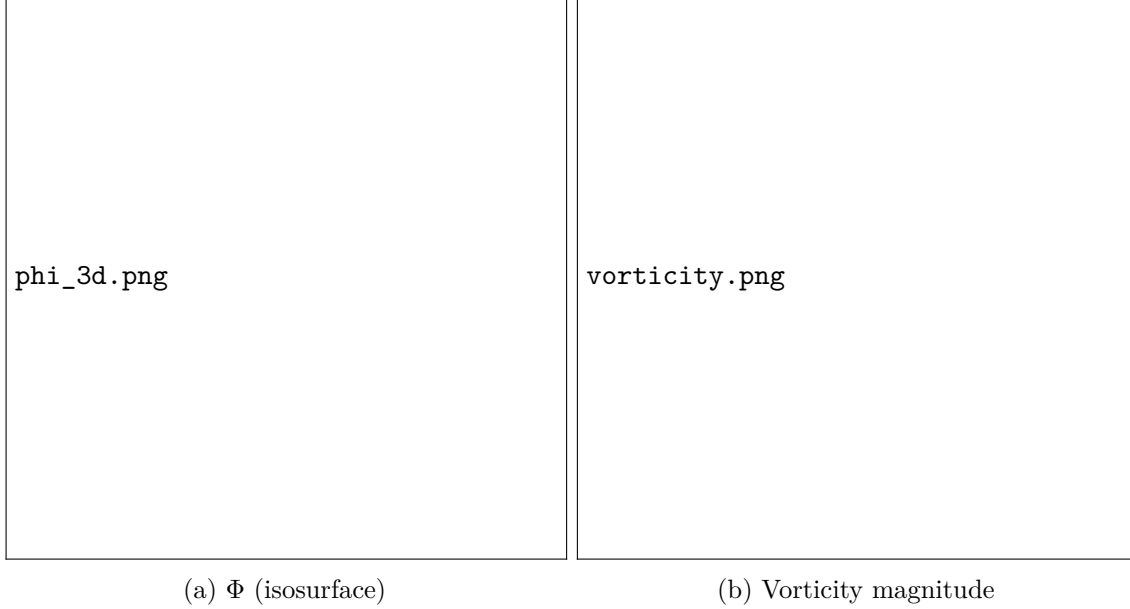


Figure 2: TPU simulation at $t = 0.1$. Void formation and rotational flow.

7 Discussion and Future Work

The RSVP stack provides a mathematically coherent, computationally scalable framework for unified science. Future directions include:

- Nonlocal kernels for long-range semantics
- Quantum field extension on Hilbert space
- Real-time inference via CLIO-trained neural fields

8 Conclusion

We have presented a unified field ontology where cosmology, cognition, and institutions emerge as scale-dependent projections of entropy-bounded dynamics. TPU-scale simulations and falsifiable predictions establish the RSVP stack as a new paradigm for physical–informational science.

A Proof of Theorem ??

Galerkin method with energy estimates...

B JAX Implementation Details

Full code available at <https://github.com/flyxion/rsvp-stack>.